# Genetic diseases

- The human genome contains about 3.2 billion DNA base pairs.
- It established that humans have only about 25,000 protein-coding genes, comprising just 1.5% of the genome.
- It is helpful to clarify three commonly used terms: hereditary, familial, and congenital.
  - Hereditary disorders: are derived from one's parents, transmitted in the gametes through the generations, and therefore are familial.
  - Congenital diseases: simply "present at birth".
- Note: some congenital diseases are not genetic (e.g., congenital syphilis). On the other hand, not all genetic diseases are congenital; the expression of Huntington disease, for example, begins only after the third or fourth decade of life.

## Nature of Genetic Abnormalities Contributing to Human Disease

There are several types of genetic abnormalities that affect the structure and function of proteins, disrupting cellular homeostasis and contributing to disease.

### a. Mutations in Protein-Coding Genes

- Mutations: permanent changes in the DNA. Those that affect germ cells are transmitted to the progeny and may give rise to inherited diseases. Mutations in somatic cells are not transmitted to the progeny but are important in the causation of cancers and some congenital malformations.

Here are some common examples of gene mutations and their effects:

1) **Point mutations**: result from the substitution of a single nucleotide base by a different base, resulting in the replacement of one amino acid by another in the protein product e.g the mutation in the β-globin chain of hemoglobin giving rise to sickle cell anemia. Such mutations are sometimes called **missense** mutations. By contrast, certain point mutations may change an amino acid codon to a chain termination codon, or a stop codon (UAG, UAA, and UGA). Such *"**nonsense**" mutations* interrupt translation, and in most cases RNAs are rapidly degraded, a phenomenon called *nonsense mediated decay*, such that little or no protein is formed.
2) **Frameshift mutation:** occur when the insertion or deletion in the genome are not in multiples of three nucleotides, and they can alter the reading frame (grouping of the codons) of the DNA strand.

3) **Trinucleotide repeat mutations**: are characterized by amplification of a sequence of three nucleotides. Although the specific nucleotide sequence that undergoes amplification varies with different disorders, all affected sequences share the nucleotides guanine (G) and cytosine (C). For example, in fragile X syndrome, there are 200 to 4000 tandem repeats of the sequence CGG within a gene called *FMR1*. In normal populations, the number of repeats is small, averaging 29. The expansions of the trinucleotide sequences prevent normal expression of the *FMR1* gene, thus giving rise to mental retardation. Another distinguishing feature of trinucleotide repeat mutations is that they are dynamic (i.e., the degree of amplification increases during gametogenesis).

**b. Alterations in Protein-Coding Genes other than mutations**

Coding genes also can undergo structural variations, such as copy number changes—*amplifications* or *deletions*—or *translocations* that result in aberrant gain or loss of protein function. As with mutations, structural changes may occur in the germline, or be acquired in somatic tissues. Pathogenic germline alterations involve a contiguous portion of a chromosome rather than a single gene, such as in the 22q microdeletion syndrome. Cancers often contain somatically acquired structural alterations, e.g. "Philadelphia chromosome"—translocation t(9;22) between the *BCR* and *ABL* genes in chronic myeloid leukemia

**c. Alterations in Non-Coding RNAs**

Recent studies indicate that a very large number of genes do not encode proteins. Instead, the nonencoded products of these genes—so-called "non-coding RNAs (ncRNAs)"—play important regulatory functions. Although many distinct families of ncRNAs exist, the two most important examples, are—small RNA molecules called microRNAs (miRNAs) and long non-coding RNAs (lncRNAs).

These can turn our attention to the following major categories of genetic disorders:

1. ***Mendelian disorders*** *resulting from mutations in single genes.* The genetic abnormalities show <u>high penetrance</u>, meaning that most individuals who inherit the anomaly show phenotypic effects. These diseases are hereditary and familial. They include many uncommon conditions, such as storage diseases and inborn errors of metabolism.
2. ***Complex disorders*** *involving multiple genes as well as environmental influences (*sometimes called multifactorial diseases) e.g. hypertension, diabetes, allergic and autoimmune diseases.

3. *Diseases arising from* ***chromosomal abnormalities****,* including changes in the number or structure of chromosomes.
4. *Other genetic diseases,* which involve single gene mutations but do not follow simple mendelian rules of inheritance. They include those resulting from triplet repeat mutations or from mutations in mitochondrial DNA, and those in which the transmission is influenced by an epigenetic phenomenon called genomic imprinting.

**Mendelian Disorders: Diseases Caused by Single-Gene Defects**

They account for approximately 1% of all adult admissions to hospitals and about 6-8% of all pediatric hospital admissions.

There are a few important tenets and caveats when considering mendelian disorders:

- Mutations involving single genes follow one of three patterns of inheritance: autosomal dominant, autosomal recessive, X-linked disease.
- A single-gene mutation may have many phenotypic effects *(pleiotropy)* and, conversely, mutations at several genetic loci may produce the same trait *(genetic heterogeneity)*. For example, Marfan syndrome, which results from a basic defect in connective tissue, is associated with widespread effects involving the skeleton, eyes, and cardiovascular system, all of which stem from a mutation in the gene encoding fibrillin. On the other hand, several different types of mutations can cause retinitis pigmentosa that associated with abnormal retinal pigmentation and consequent visual impairment.
- The phenotypic manifestations of mutations affecting a known single gene are influenced by other genetic loci (modifier genes). e.g. in cystic fibrosis, these modifier loci can affect the severity or extent of the disease.
- The use of proactive prenatal genetic screening in high-risk populations has significantly reduced the incidence of certain genetic disorders.

Table: Estimated Prevalence of Selected Mendelian Disorders Among Live-Born Infants.

| Disorder | Estimated Prevalence |
|---|---|
| **Autosomal Dominant Inheritance** | |
| Familial hypercholesterolemia | 1 in 500 |
| Polycystic kidney disease | 1 in 1000 |
| Hereditary spherocytosis | 1 in 5000 (northern Europe) |
| Marfan syndrome | 1 in 5000 |
| Huntington disease | 1 in 10,000 |
| **Autosomal Recessive Inheritance** | |
| Sickle cell anemia | 1 in 500 (U.S. African Americans)* |
| Cystic fibrosis | 1 in 3200 (U.S. Caucasians) |
| Tay-Sachs disease | 1 in 3500 (U.S. Ashkenazi Jewish; French Canadians) |
| Phenylketonuria | 1 in 10,000 |
| MPSs—all types | 1 in 25,000 |
| Glycogen storage diseases—all types | 1 in 50,000 |
| Galactosemia | 1 in 60,000 |
| **X-Linked Inheritance** | |
| Duchenne muscular dystrophy | 1 in 3500 (U.S. males) |
| Hemophilia | 1 in 5000 (U.S. males) |

*MPS*, Mucopolysaccharidosis.

Table: Biochemical Basis and Inheritance Pattern for Selected Mendelian Disorders.

| Disease | Abnormal Protein | Protein Type/Function |
|---|---|---|
| **Autosomal Dominant Inheritance** | | |
| Familial hypercholesterolemia | LDL receptor | Receptor transport |
| Marfan syndrome | Fibrillin | Structural support: extracellular matrix |
| Ehler-Danlos syndrome* | Collagen | Structural support: extracellular matrix |
| Hereditary spherocytosis | Spectrin, ankyrin, or protein 4.1 | Structural support: red blood cell membrane |
| Neurofibromatosis, type 1 | Neurofibromin-1 (NF-1) | Growth regulation |
| Adult polycystic kidney disease | Polycystin-1 (PKD-1) | Cell–cell and cell–matrix interactions |
| **Autosomal Recessive Inheritance** | | |
| Cystic fibrosis | Cystic fibrosis transmembrane regulator | Ion channel |
| Phenylketonuria | Phenylalanine hydroxylase | Enzyme |
| Tay-Sachs disease | Hexosaminidase | Enzyme |
| Severe combined immunodeficiency | Adenosine deaminase | Enzyme |
| α- and β-thalassemias† | Hemoglobin | Oxygen transport |
| Sickle cell anemia† | Hemoglobin | Oxygen transport |
| **X-Linked Recessive Inheritance** | | |
| Hemophilia A | Factor VIII | Coagulation |
| Duchenne/Becker muscular dystrophy | Dystrophin | Structural support: cell membrane |
| Fragile X syndrome | FMRP | RNA translation |

## Transmission Patterns of Single-Gene Disorders

### Disorders of Autosomal Dominant Inheritance

1. They are manifested in the **heterozygous state**, so at least one parent in an index case usually is affected. Both males and females can be affected,
2. Both sexes can transmit the condition and are affected.
3. When an affected person marries an unaffected one, each child has 50% chance of having the disease.

**N***N--------NN

| **N***N | **N***N | NN | NN |
|---|---|---|---|
| 25% affected | 25% affected | normal | normal |
| 50% | | 50% | |

4. With any autosomal dominant disorder, some patients do not have affected parents. Such patients owe their disorder to new mutations involving either the egg or the sperm from which they were derived. Their siblings are neither affected nor at increased risk for development of the disease.
5. Clinical features can be modified by reduced penetrance and variable expressivity.

- ***Reduced penetrance*:** Some persons inherit the mutant gene but are phenotypically normal.
- The phenomenon is called ***variable expressivity***, if a trait is consistently associated with a mutant gene but is expressed differently among persons carrying the gene. e.g. manifestations of neurofibromatosis 1 range from brownish spots on the skin to multiple tumors and skeletal deformities.

6. In many conditions, the age at onset is delayed, and symptoms and signs do not appear until adulthood (as in Huntington disease).
7. A 50% reduction in the normal gene product is associated with clinical signs and symptoms.
8. Because a 50% loss of enzyme activity can be compensated for, involved genes in autosomal dominant disorders usually do not encode enzyme proteins, but instead fall into two other categories of proteins:

a. Those involved in regulation of complex metabolic pathways, (e.g. familial hypercholesterolemia, results from mutation in the (LDL) receptor gene).

b. Key structural proteins, such as collagen and components of the red cell membrane skeleton (e.g., spectrin, abnormalities of which result in hereditary spherocytosis).

**Examples of autosomal dominant disorders:**

Marfan syndrome

Hypercholesterolemia

Polycystic kidney disease

Hereditary spherocytosis

Familial polyposis coli

**Disorders of Autosomal Recessive Inheritance**

They are manifested in the **homozygous state**. They occur when both of the alleles at a given gene locus are mutants. Therefore, such disorders are characterized by the following features:

1. The trait does not usually affect the parents, who are carriers of one diseased allele, but multiple siblings may show the disease.
2. Siblings have 25% chance are being affected (i.e., the recurrence risk is 25% for each birth).

| | | | |
|---|---|---|---|
| | **N***N | **N***N | |
| **N*N*** | **N***N | **N***N | NN |
| affected | carrier | carrier | normal |
| 25% | 25% | 25% | 25% |

3. if the mutant gene occurs with a low frequency in the population, there is a strong likelihood that the affected patient (the proband) is the product of a consanguineous marriage.
4. The expression of the defect tends to be more uniform than in autosomal dominant disorders.
5. Complete penetrance is common.
6. Onset is frequently early in life.
7. Although new mutations for recessive disorders do occur, they are rarely detected clinically. Because the affected person is an asymptomatic heterozygote, several generations may pass before the descendants of such a person mate with other heterozygotes and produce affected offspring.

8. In many cases, enzymes are affected by the mutation. In heterozygotes, equal amounts of normal and defective enzyme are synthesized. Usually the natural "margin of safety" ensures that cells with half of their complement of the enzyme function normally.

**Examples of autosomal recessive disorders:**

Sickle cell anemia
Thalassemia
Cystic fibrosis
Phenylketonuria
Wilson disease
Glycogen storage diseases
Albinism

**X-Linked Disorders** (most are X-linked recessive)

For the most part, **sex-linked disorders are X-linked recessive,** and are characterized by:

1. Heterozygous female carriers transmit them only to sons, who are hemizygous for the X chromosome.
2. Heterozygous females rarely express the full phenotypic change, because they have the paired normal allele. Although one of the X chromosomes in females is inactivated, this process of inactivation is random, which typically allows sufficient numbers of cells with the normal expressed allele to emerge.
3. An affected male does not transmit the disorder to sons, but all daughters are carriers.

XX **X***Y

**X***X XY **X***X XY

All daughters are carrier, sons not affected

4. Sons of heterozygous women have 50% chance of receiving the mutant gene (being affected).

**X***X XY

**X***X **X***Y XX XY

Sons are 50% affected

5. There are few X-linked dominant (e.g. Vitamin D-resistant rickets) and their inherited pattern characterized by transmission of the dis. to 50% of sons and daughters of an affected heterozygous female.
6. In X-linked dominant, an affected male cannot transmit the dis. to his sons but all daughters are affected.

**Examples of X-linked disorders:**

Hemophilia A and B

Glucose-6-phosphate dehydrogenase deficiency (G6PD deficiency)

Agammaglobulinemia

Duchenne muscular dystrophy

## Diseases Caused by Mutations in Genes Encoding Structural Proteins

## Marfan Syndrome

Is an **autosomal dominant disorder** of connective tissues (inherited defect in an extracellular glycoprotein called **fibrillin-1**), both quantitative & qualitative defects have been noted.

- This glycoprotein, secreted by fibroblasts, is the major component of microfibrils found in the extracellular matrix. Microfibrils serve as scaffolding for the deposition of tropoelastin, an integral component of elastic fibers. They are particularly abundant in the aorta, ligaments, and the ciliary zonules that support the ocular lens, precisely the tissues that are prominently affected in Marfan syndrome.
- Morphology: 3 systems are mainly affected:

**Skeletal abnormalities*: Slender patient with abnormally long legs, arms & fingers (arachnodactyly); high arched palate, hyperextensibility of joints. Also, a variety of spinal deformities (kyphoscoliosis) & the chest shows pectus excavatum or a pigeon-breast deformity.

* *Eyes:* bilateral dislocation or subluxation of the lens due to weakness of suspensory ligament **(ectopia lentis).**

* *Cardiovascular system*: Fragmentation of the elastic fibers in the tunica media of the aorta → aneurysmal dilatation & aortic dissection. Loss of medial support causes dilatation of aortic valve ring → aortic incompetence. The cardiac valves, especially the mitral valve, may be excessively distensible and regurgitant **(floppy valve syndrome),** → mitral valve prolapse and congestive heart failure.

Aortic rupture is the most common cause of death and may occur at any age.

*Variable expression of the features above between different patients.

**Diseases Caused by Mutations in Genes Encoding Receptor Proteins or Channels**

**Familial Hypercholesterolemia**

1. Autosomal dominant disease. Heterozygotes have 2-3 folds elevation of plasma cholesterol levels. Although their cholesterol levels are elevated from birth, heterozygotes remain asymptomatic until adulthood when they develop xanthoma along tendon sheaths & premature atherosclerosis resulting in coronary artery disease.
2. While homozygotes are much more severely affected, developing cutaneous xanthoma in childhood & dying from myocardial infarction before the age of 20 years.
3. It is caused by mutation in the gene that specifies the receptor for low density lipoproteins LDL.
4. Cholesterol may be derived from diet or from endogenous synthesis;

Normally, the endogenous synthesis of cholesterol and LDL begins in the liver. The first step in the synthesis is the secretion of triglyceride-rich very-low-density lipoprotein (VLDL) by the liver into the blood. In the capillaries of adipose tissue and muscle, the VLDL particle undergoes lipolysis and is converted to intermediate-density lipoprotein (IDL). In comparison with VLDL, the content of triglyceride is reduced and that of cholesteryl esters is enriched in IDL, but IDL retains on its surface the VLDL-associated apolipoproteins B-100 and E. Further metabolism of IDL occurs along two pathways: Most of the IDL particles are directly taken up by the liver through the LDL receptor; others are converted to cholesterol-rich LDL by a further loss of triglycerides and the loss of apolipoprotein E. In the liver cells, IDL is recycled to generate VLDL.

Mutation in LDL receptor gene---accumulation of LDL cholesterol in the plasma, in addition the absence of LDL receptors on the liver impair the transport of IDL to the liver---accumulation of IDL that converted to LDL.

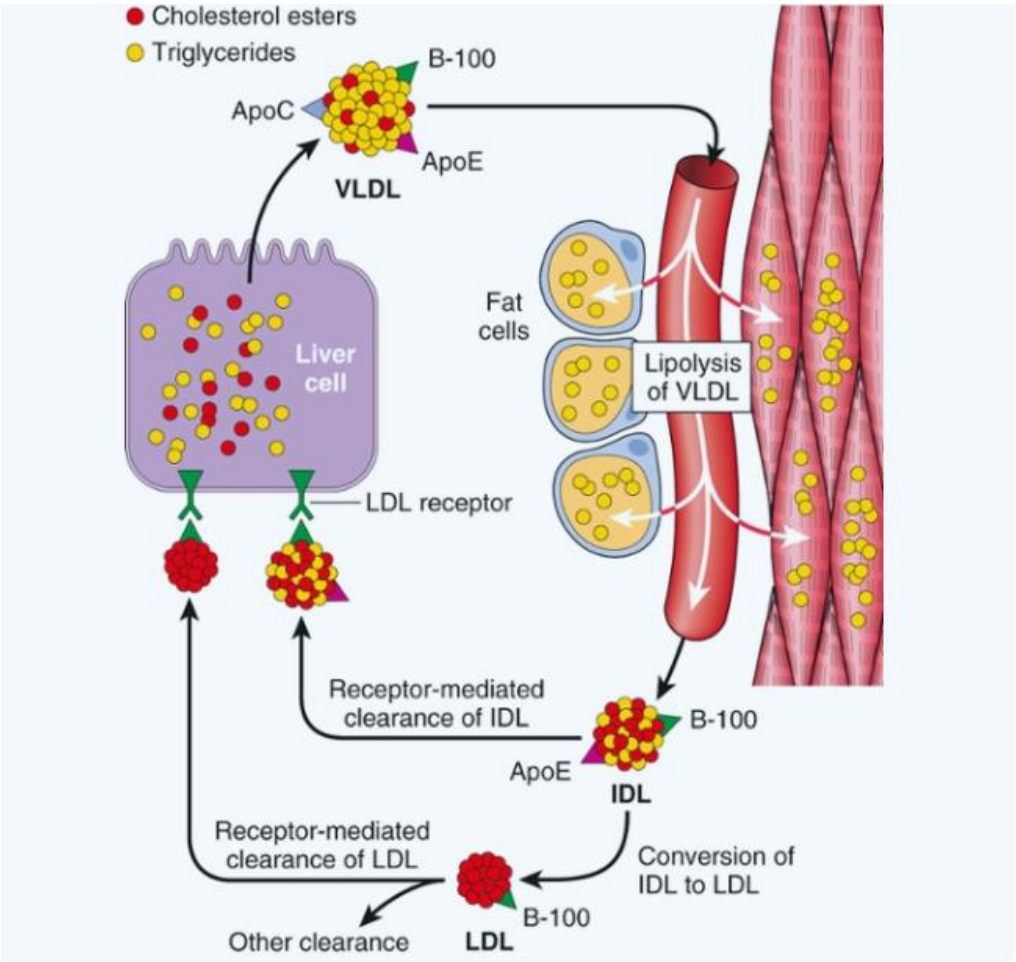


**Fig Low-density lipoprotein (LDL) metabolism and the role of the liver in its synthesis and clearance. Lipolysis of very-low-density lipoprotein (VLDL) by lipoprotein lipase in the capillaries releases triglycerides, which are then stored in fat cells and used as a source of energy in skeletal muscles. IDL (intermediate-density lipoprotein) remains in the blood and is taken up by the liver.**